

WEEK 4 TECHNICAL REPORT

Privacy architecture, local SLM latency, and dependency gate

Benchmark run: 29 Aug 2026 | Evidence rechecked: 1 Sep 2026

Outcome: All three assigned Week 4 deliverables are complete. Overall system privacy remains a conditional pass until the application SLM client and disconnected-network test exist.

4	2.22 s	1 rule
PRIVACY PRINCIPLE AREAS	MEAN LOCAL LATENCY	DEPENDENCY CHECK IN EVERY PR

Week 4 scope

- Draft privacy/architecture principles for network calls, telemetry, logs, and dependencies.
- Run and confirm a real-machine SLM latency benchmark using the selected local model.
- Set the standing rule that every new dependency receives a quick privacy check in its PR description.

How to read the evidence

Evidence class	What is supported	Status
Documented	A written design rule or team policy exists.	Complete
Measured	A local benchmark produced recorded values on this Mac.	Measured
Repository checked	Source, manifests, logs, tests, and runtime evidence were reviewed.	Verified
End-to-end	The final integrated app has not yet been tested with public networking disabled.	Pending

1. Privacy architecture principles

Design target, not a claim that integration is finished. Participant data and dataset-derived features should remain on the local machine. Sensitive runtime paths may use loopback services such as Ollama, while public network access is denied by default.

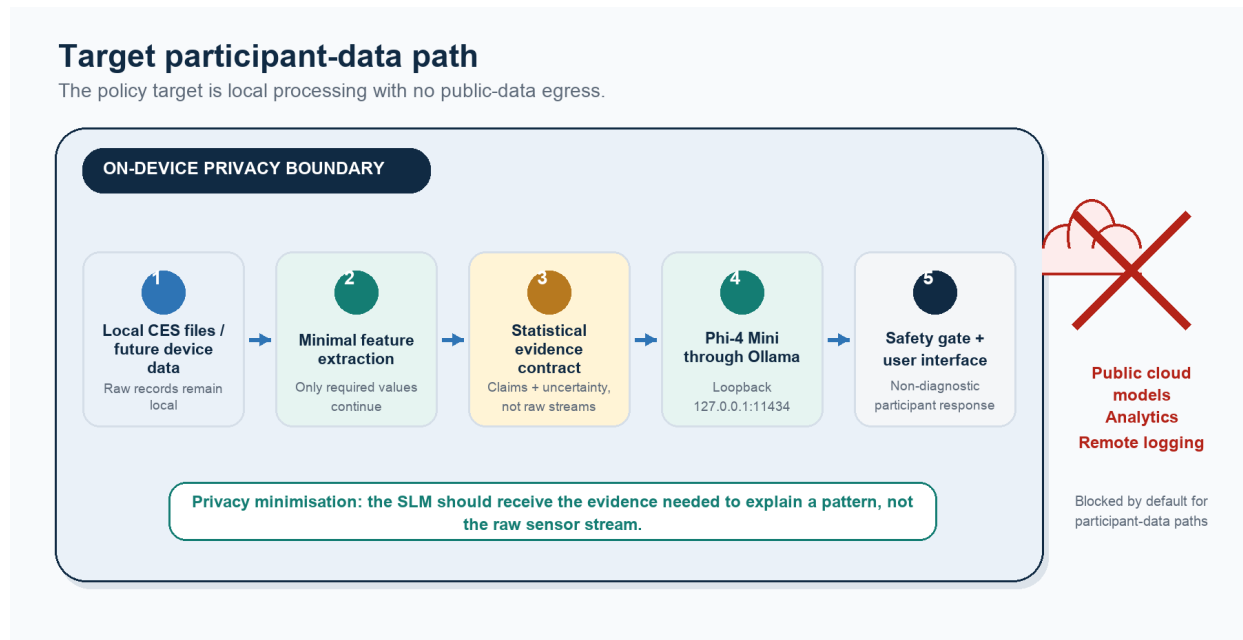


Figure 1. Target local-first participant-data path and public-network boundary.

Core principles

- **Network:** participant-data processing, inference, logging, and safety paths must not call public endpoints by default.
- **Telemetry:** analytics, tracking, crash uploads, session replay, and remote performance monitoring are disabled by default.
- **Logs:** retain aggregate technical evidence only; exclude raw sensors, participant IDs, free text, prompts, responses, PHQ-4 values, and crisis text.
- **Dependencies:** each package, model runtime, install script, and downloaded binary is part of the privacy boundary.

Important distinction: A localhost endpoint proves that the benchmark request targeted this machine. It does not by itself prove that the complete future app works offline. That claim needs an integrated client plus a Wi-Fi-off or public-network-blocked test.

2. Network calls, telemetry, and logs

Four controls define the privacy baseline

Each control has a default rule and a current repository finding.

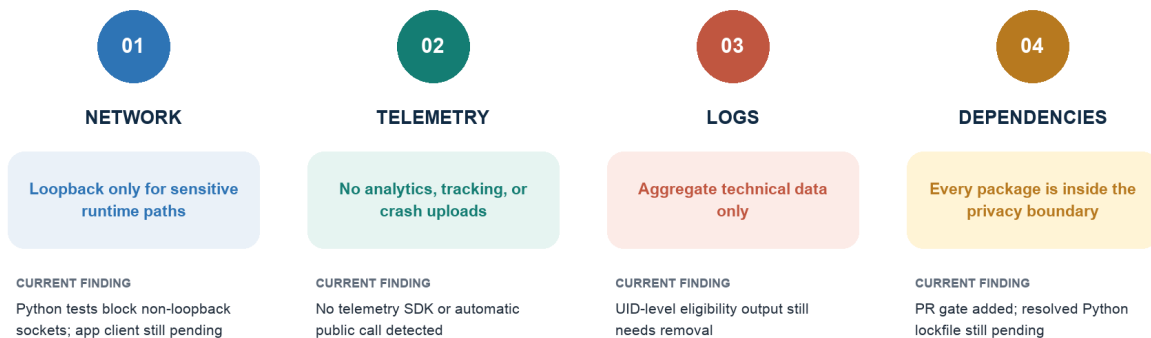


Figure 2. The written rules and the current repository evidence are tracked separately.

Network decision rule

Category	Decision
Allowed by default	Loopback communication between local components; development downloads before participant data is present.
Privacy review required	Any public endpoint, cloud model, CDN or remote asset, telemetry SDK, or install-time model/binary download.
Blocked by default	Outbound public calls from participant-data processing, inference, logging, or safety paths.

Logging boundary

Allowed technical evidence	Sensitive content that must not be logged
Aggregate latency, model tag, component events, non-identifying error class, privacy-safe record counts	Raw sensing records, participant IDs, free text, prompts/responses linked to a person, wellbeing scores, crisis text, identifying paths

Repository findings checked on 1 Sep 2026

- No telemetry, analytics, crash-reporting, cloud-model, CDN, or automatic public frontend call was detected in the reviewed application paths.
- The frontend still contains six clickable starter links to public sites. They are user-initiated rather than automatic calls, but must be removed or approved before participant-facing use.
- The benchmark uses Python urllib only for the loopback Ollama endpoint.
- backend/data_pipeline/ces_eligibility.py still emits ineligible_uids and UID-keyed reasons; shared logs or screenshots must use aggregate reason counts instead.

3. Standing dependency privacy rule

Every PR that adds or materially changes a dependency must include a completed 10-minute privacy spot-check in the PR description. The repository PR template now contains that section.

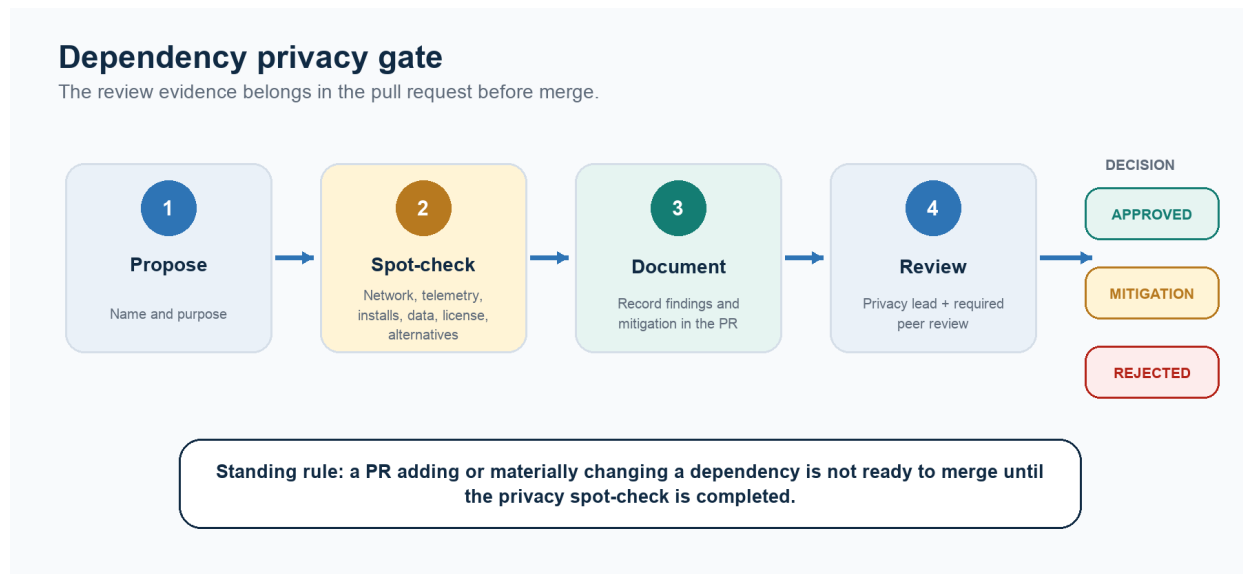


Figure 3. The privacy evidence is recorded before the merge decision.

The six required checks

- Default network calls or phone-home behaviour.
- Telemetry, analytics, crash reports, diagnostics, or usage metrics.
- Install scripts, post-install downloads, auto-updates, or remote model/binary downloads.
- Access to participant data, prompts, SLM responses, wellbeing labels, or derived features.
- License suitability and current maintenance status.
- Whether the standard library or an existing dependency can do the job.

Current limitation: Earlier dependency additions were committed without this PR check. A retrospective audit was completed, but Python declarations remain mostly unpinned and no resolved Python lockfile exists yet.

4. Confirmed SLM latency benchmark

Phi-4 Mini was benchmarked through the local Ollama API using five synthetic prompts. No participant data was used. The output file recorded all five runs successfully.

Local Phi-4 Mini latency

Five synthetic prompts through Ollama on the benchmark Mac

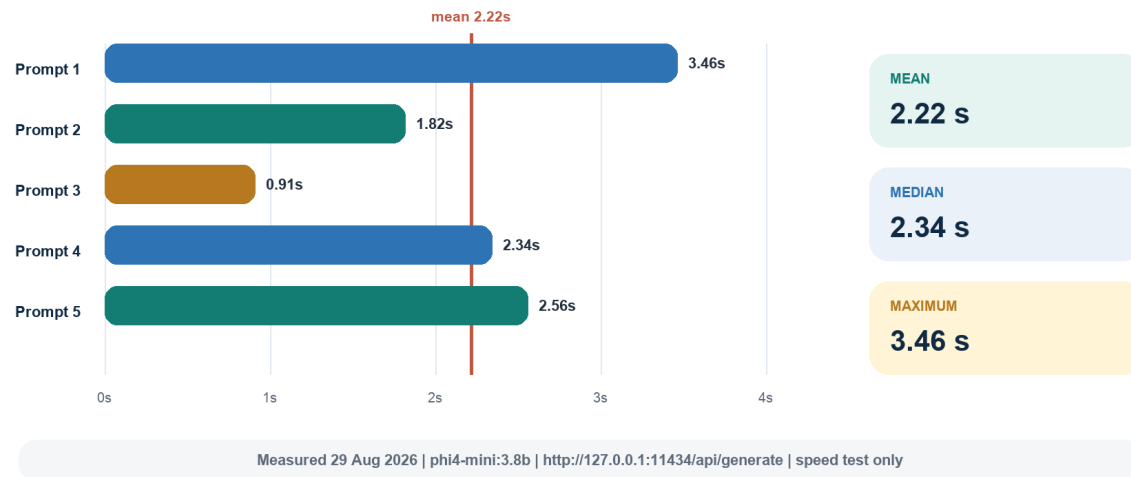


Figure 4. Recorded real-machine response latency for five synthetic prompts.

Benchmark field	Recorded result
Provider / model	Ollama / phi4-mini:3.8b
Endpoint	http://127.0.0.1:11434/api/generate
Machine	Apple arm64 Mac; Python 3.14.0; benchmark timestamp 29 Aug 2026 13:01 AEST
Runs	5 synthetic prompts; 5 successful responses
Latency	min 0.91s mean 2.22s median 2.34s sample p95 3.28s max 3.46s

Correct interpretation

- The measured mean of 2.22 seconds supports continued prototype work on this machine; it is not a universal performance guarantee.
- Five runs are a small sample. The recorded p95 is an interpolated sample statistic, not a production-grade tail-latency estimate.
- This benchmark measures speed only. It does not establish clinical validity, response safety, privacy of the final integrated app, or crisis-handling quality.
- One raw response used causal-sounding wording, so all product output still needs a deterministic claim and safety gate before display.

5. Evidence, conclusions, and follow-ups

Evidence class	What is supported	Status
Week 4 assignment	Principles drafted, local benchmark recorded, and dependency rule added to the PR template.	Complete
Repository privacy	No automatic cloud integration or telemetry detected; loopback-only Python socket policy exists.	Conditional
Local SLM speed	Mean 2.22 seconds across five synthetic prompts using phi4-mini:3.8b.	Measured
Offline end-to-end	Application client is absent and public-network-disabled inference has not been tested.	Pending

What is known

- The selected local model is Ollama with the exact tag phi4-mini:3.8b; the tag is recorded in the model manifest.
- The CES dataset and local virtual environment are not tracked in Git or present in the reviewed Git history.
- The recorded Week 4 verification passed 124 Python tests under the loopback-only socket policy, plus frontend lint and production build checks.
- This document was rechecked against branch yuktha/privacy-week4 at commit 517ade3.

Follow-ups before participant-facing use

- Replace UID-level eligibility output with aggregate reason counts.
- Remove or explicitly approve the six frontend starter links.
- Create and use a resolved Python dependency lockfile.
- Build the actual backend SLM client and keep it loopback-only.
- Run an integrated Ollama inference test with public networking disabled.
- Enforce the dependency privacy spot-check on every future dependency PR.

Repository evidence used

Privacy design: privacy/privacy_architecture_principles.md and privacy/dependency_privacy_checklist.md | PR rule: .github/pull_request_template.md | Benchmark: benchmarks/slm_latency_benchmark.py and benchmarks/slm_latency_results.json | Verification: privacy/initial_dependency_audit.md, privacy/week4_privacy_status.md, tests/privacy/test_no_network_egress.py, and pytest.ini

Short explanation for the team

Team summary: My Week 4 work established MindSense's local-first privacy baseline, measured Phi-4 Mini at 2.22 seconds mean latency across five synthetic prompts, and added the dependency PR spot-check. The assignment is complete. Final offline end-to-end assurance remains pending the application client and disconnected-network test.